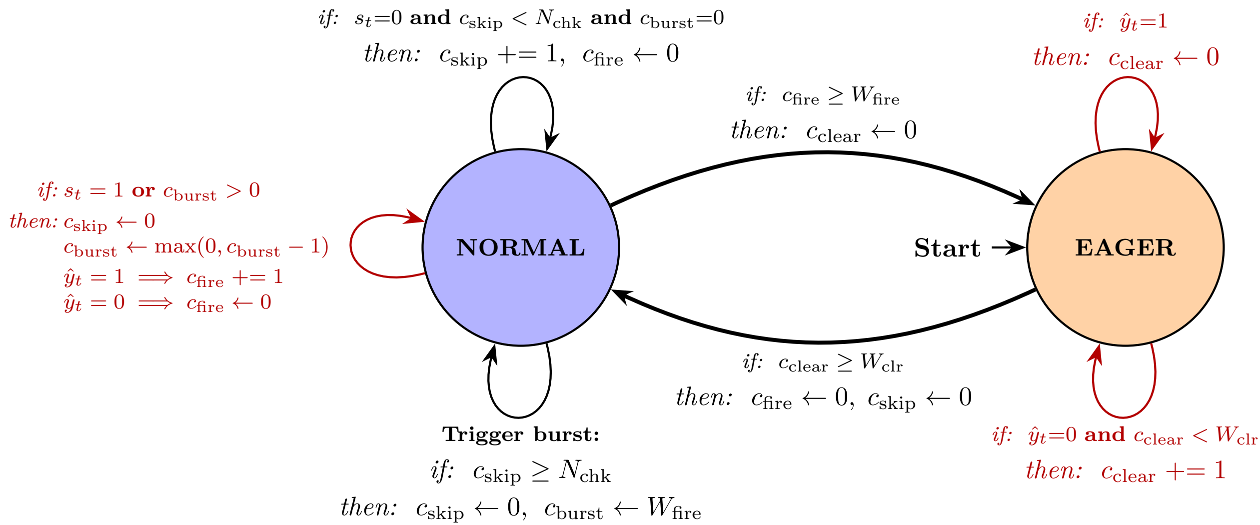

Operational Modes

NORMAL	**Skip Module enabled.** Model $\mathcal{M}$ is invoked only if motion is detected ($s_t = 1$) or a forced burst is active.
EAGER	**Continuous inference.** Skip module $\mathcal{S}$ is bypassed; $\mathcal{M}$ evaluates every frame. (Default starting state).

Transition Types

(red arrow)	**Inference executed:** $\mathcal{M}(f_t)$ is invoked to evaluate the current frame.
(black arrow)	**Zero-inference step:** Frame is skipped, or state variables are updated seamlessly.

Variables & Parameters

$s_t \in \{0, 1\}$	Motion sensor ($\mathcal{S}$) prediction (1: motion, 0: static).
$\hat{y}_t \in \{0, 1\}$	Fire model ($\mathcal{M}$) prediction (1: fire/smoke, 0: safe).
c_{fire}	Counter for consecutive positive detections ($\hat{y}_t = 1$).
c_{clear}	Counter for consecutive safe frames ($\hat{y}_t = 0$) during EAGER mode.
c_{skip}	Counter for consecutively skipped frames during NORMAL mode.
c_{burst}	Countdown variable for remaining frames in a forced inference burst.
W_{fire}	Threshold of consecutive positive detections required to trigger EAGER mode.
W_{clr}	Threshold of consecutive negative frames required to return to NORMAL mode.
N_{chk}	Maximum allowed skipped frames before triggering a forced burst.